VU Quiz Collection

Total Questions: 19 from 19 Quiz(es)

Courses: MTH501

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Quiz 1 - MTH501 (Q1/10)

Question:

If {u1,u2,…,up}{u1,u2,…,up}\{u_{1}, u_{2}, …, u_{p}\} is an orthonormal basis for a subspace WWW of RnRnR^n, then ...

Options:

A. Projwy=(y.u1)up+(y.u2)up+…+(y.up)upProjwy=(y.u1)up+(y.u2)up+…+(y.up)upProj_{w}y=(y.u_{1})u_{p}+(y.u_{2})u_{p}+…+(y.u_{p})u_{p}

B. Projwy=(y.u1)u1+(y.u2)u2+…+(y.up)upProjwy=(y.u1)u1+(y.u2)u2+…+(y.up)upProj_{w}y=(y.u_{1})u_{1}+(y.u_{2})u_{2}+…+(y.u_{p})u_{p}

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Quiz 2 - MTH501 (Q2/10)

Question:

Two vectors are _____if at least one of the vector is a multiple of the other

Options:

A. linearly independent

B. linearly dependent

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Quiz 3 - MTH501 (Q3/10)

Question:

Two vectors uuu and vvv in RnRnR^n are orthogonal if ------

Options:

A. u.v=0u.v=0u.v=0

B. u . v"` 0 u . v"` 0 u . v \n e q 0

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Quiz 4 - MTH501 (Q4/10)

Question:

A unit vector is obtained by ------

Options:

A. $u={||v||}/{||v||}

B. $u=v/{||v||}

C. None of the above

D. $u={||v||}/v

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Quiz 5 - MTH501 (Q5/10)

Question:

Let W be a subspace of RnRn{R^n} and {u1,u2,...,up}{u1,u2,...,up}\{ {u_1},{u_2},...,{u_p}\} is any orthogonal basis of W, then
y\^=c1u1+c2u2+...+cnuny\^=c1u1+c2u2+...+cnun\mathop y\limits^\^ = {c_1}{u_1} + {c_2}{u_2} + ... + {c_n}{u_n}
Where

Options:

A. cj=y.ujuj.ujcj=y.ujuj.uj{c_j} = \frac{{y.{u_j}}}{{{u_j}.{u_j}}}

B. cj=y.ujycj=y.ujy{c_j} = \frac{{y.{u_j}}}{y}

C. None of these

D. cj=y.ujujcj=y.ujuj{c_j} = \frac{{y.{u_j}}}{{u_j}}

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Quiz 6 - MTH501 (Q6/10)

Question:

Elementary row operations on a matrix do not affect the -------- relations among the columns of the matrix.

Options:

A. Linear independence

B. Linear dependence

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Quiz 7 - MTH501 (Q1/10)

Question:

The matrix A^T(Transpose of A) x A is _____ if and only if the columns of A are linearly independent.

Options:

A. singular

B. scalar

C. symmetric

D. invertible

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Quiz 8 - MTH501 (Q8/10)

Question:

If a matrix U has orthonormal columns, then UUT (transpose of U) = _____.

Options:

A. negative

B. I

C. 0

D. positive

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Quiz 9 - MTH501 (Q9/10)

Question:

Which of the following set of vectors is NOT an orthogonal set?

Options:

A. (1, 2, 0) ;(1, 0, 3)

B. (1, 1, 1) ;(1, 0,-1)

C. (2, 3) ;(-6, 4)

D. (3, 0, 0) ;(0, 1, 0)

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Quiz 10 - MTH501 (Q10/10)

Question:

An n x n matrix A is invertible if and only if _____ is not an eigen value of A.

Options:

A. 0

B. -1

C. 1

D. 2

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Quiz 11 - MTH501 (Q2/10)

Question:

Let s={u1+u2+.....up}s={u1+u2+.....up}s = \{ {u_1} + {u_2} + ...{u_p}\} be a basis for a subspace W of RnRn{R^n} , is also an orthogonal basis if S is an orthogonal set.

Options:

A. True

B. False

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Quiz 12 - MTH501 (Q3/10)

Question:

A square matrix with _____ columns is an orthogonal matrix.

Options:

A. orthogonal

B. orthonormal

C. even

D. same

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Quiz 13 - MTH501 (Q4/10)

Question:

A m ×n matrix U has orthonormal columns if and only if UtU=IUtU=I{U^t}U = I

Options:

A. True

B. False

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Quiz 14 - MTH501 (Q5/10)

Question:

If x is _____ to both u and v, then x must be orthogonal to u – v.

Options:

A. orthogonal

B. orthonormal

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Quiz 15 - MTH501 (Q6/10)

Question:

The product of upper triangular matrices is:

Options:

A. Diagonal matrix.

B. Upper triangular matrix.

C. Lower triangular matrix

D. None of the above

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Quiz 16 - MTH501 (Q7/10)

Question:

Let W be a subspace of RnRnR^n, y be any vector in RnRnR^n and yˆy^\widehat y the orthogonal projection of yyy onto WWW. Then yˆy^\widehat y is the closest point in WWW to yyy, in the sense that

Options:

A. ||y" yˆ||<||y" v||||y" y^||<||y" v||||y-\widehat y||<||y-v||

B. ||y" yˆ||>||y" v||||y" y^||>||y" v||||y-\widehat y ||>||y-v||

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Quiz 17 - MTH501 (Q8/10)

Question:

The matrix A^T(Transpose of A) x A is invertible if and only if the columns of A are linearly independent.

Options:

A. TRUE

B. FALSE

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Quiz 18 - MTH501 (Q9/10)

Question:

A partitioned square matrix ‘A’ is said to be _____ matrix if the matrices on the main diagonal are square and all matrices above the main diagonal are zero.

Options:

A. Diagonal-constant

B. Null

C. Block upper triangular

D. Block lower triangular

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Quiz 19 - MTH501 (Q10/10)

Question:

If we divide a non-zero vector by its length we get a

Options:

A. normalized vector

B. none of these

C. unit vector

D. zero vector

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